

QRCx: Sequential Dissipative Quantum Reservoir Computing for Operational Weather Forecasting

QRCx Team

qBraid $\times$ MITRE $\times$ JonesTrading Global Industry Challenge 2026, Phase 3, Track B

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1 Track and Rationale

We build a sequential dissipative quantum reservoir computer (QRC) for real hourly weather forecasting at NOAA station KORD (Chicago O’Hare), held to a fairness protocol no reviewer can attack: identical features, identical strict temporal splits, and a dimension-matched classical echo-state network (ESN) plus a null-control linear baseline under the exact same protocol. Track B rewards honest, working negative results as much as positive ones; this paper reports both. Our core question is narrow and falsifiable: *does engineered dissipation, not just unitary evolution, provide the reservoir’s forecasting-relevant memory?* We answer yes, with a statistically significant ablation (Sec. 4.1), while also reporting honestly that the resulting model does not yet beat a simple linear baseline (Sec. 4.3) – a finding the fairness protocol was built to surface, not hide.

2 Architecture

Figure 1 shows the pipeline. A real hourly 13-feature vector x_k (dry-bulb and dew-point temperature, relative humidity, wind speed/direction decomposed into W_x, W_y , sea-level pressure, temperature depression, and four cyclical hour-of-day/day-of-year encodings) is injected via single-qubit RY rotations (or, in an alternative encoding mode, a ZZ-feature-map-style layer) into a transverse-field Ising model (TFIM) reservoir of N qubits, whose density matrix ρ_k evolves coherently under the exact propagator $e^{-iH\tau}$ (eigendecomposition-based, not Trotterized) before a dissipative channel D_{γ_1, γ_2} – amplitude damping (γ_1) and dephasing (γ_2) – is applied, and *persists to the next timestep*: this is a genuinely recurrent reservoir, not a fresh-encode-per-sample windowed one. A time-multiplexed readout takes V equally-spaced intermediate snapshots of single- and two-body Pauli correlators per coherent-evolution interval (non-destructively; ρ keeps evolving), giving $3N + 3\binom{N}{2}$ correlators per sub-step (234 at the $N = 12$ reference configuration). A Ridge regression readout predicts the climatological-anomaly residual target, either directly or as a delta added back to persistence (“residual” architecture). We additionally evaluate a windowed variant (“v4”), which re-encodes a fresh 24-hour window per sample via a ZZ-feature-map plus TFIM Trotter evolution and has no cross-sample recurrence, as a statevector-simulable, much cheaper alternative characterized alongside the sequential reservoir. **These are two different architectures with different simulation costs, not two measurements of one config:** v5 (the reference architecture) is a 12-qubit density-matrix simulation; v4 (a supplementary, computationally cheaper variant) is a 20-qubit statevector simulation, and neither number should be read as a scaling curve of the other. v4’s real circuit depth/gate count, measured via PennyLane’s `qml.specs` on the exact encoding+evolution circuit (not a hand estimate): at the 12-qubit reference dimension, depth 191 (1,170 gates: 858 IsingZZ, 156 RZ, 120 RX, 36 Hadamard); at v4’s own native

20-qubit config, depth 303 (2,990 gates). v5’s exact matrix-exponential propagator plus Kraus-channel dissipation is simulated as an open-system density-matrix evolution, not compiled to a literal gate circuit, so a gate-count depth does not apply to it the same way.

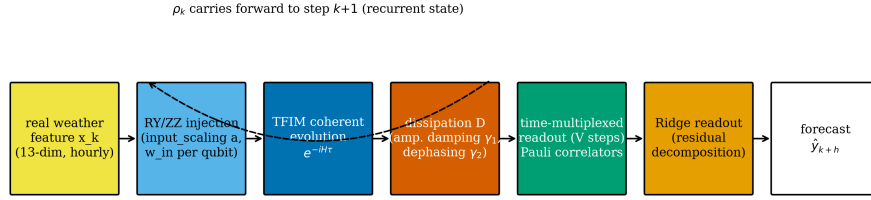


Figure 1: Sequential dissipative reservoir pipeline. ρ_k (dashed feedback) carries the recurrent quantum state from step k to $k + 1$ – the architectural feature this paper’s central ablation (Sec. 4.2) isolates.

3 IPC-Matched Tuning (Novel Contribution)

Reservoir hyperparameters are conventionally tuned to maximize *total* Information Processing Capacity (IPC; Dambre et al. 2012, extended to dissipative reservoirs by Čindrak et al. 2026) – but a forecasting task only benefits from capacity in the specific (delay, degree) region it actually uses. We introduce a **task demand profile** $D(\text{delay, degree})$: how much of the real KORD residual target’s h -step-ahead value is explained by a degree-1/2/3 Legendre polynomial of its own lagged history, at the same (delay, degree) resolution as the reservoir’s supply-side IPC $C(\text{lag, degree})$ (extended here to degree 3 via probabilists’ Hermite polynomials of the reservoir’s iid-Gaussian drive). Matching – maximizing captured capacity $\sum_h \sum \min(C, D)$ over a coarse (γ_1, a) grid rather than raw $\sum C$ – selected $\gamma_1=0.3, a=0.1$ (captured capacity 9.26, vs. 4.53 for the prior reference config: a +104% overlap improvement). **Honest result**: on a real (reduced-sample) pilot forecast comparison, the matched configuration improved skill at $h=1, 3$ but the reference configuration remained better at $h=6$ (this task’s own named target horizon) and $h=12$ (Table 1) – maximizing univariate self-demand capture does not guarantee improving the true multivariate-readout forecast task. We report this as a genuine, unresolved limitation rather than omit or reframe it.

Config	skill@1h	skill@3h	skill@6h	skill@12h
Matched ($\gamma_1=0.3, a=0.1$)	−1.72%	−3.82%	−4.12%	+8.89%
Reference ($\gamma_1=0.03, a=0.3$)	−4.20%	−4.59%	−1.22%	+11.15%

Table 1: IPC-matched vs. reference configuration, real reduced-sample pilot forecast skill vs. persistence.

4 Results

4.1 Dissipation Is the Computational Mechanism

The central architectural claim – that engineered dissipation, not mere unitary evolution, provides the reservoir’s task-relevant memory – is tested directly by comparing the tuned configuration ($\gamma_1=0.03$) against an otherwise-identical dissipation-off diagnostic ($\gamma_1=0$), same seed, same real pilot data, 12-qubit reference config (Table 2).

V1 (paradigm validation): pass. **V2 (competitive verdict): fail** – the same +14.4% result loses to dimension-matched ESN at 3 of 4 tested horizons (Sec. 4.3).

γ_1	skill@1h	skill@3h	skill@6h	skill@12h (DM p)
0.03 (tuned, residual)	−3.51%	−1.36%	+5.79%	+ 14.39% ($p \approx 0$)
0.0 (off, residual)	−0.01%	−0.01%	−0.03%	−0.06% ($p=0.16$)

Table 2: Dissipation-on vs. off ablation, same seed and data. With dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at every horizon; with it on, a real, Diebold–Mariano-significant effect emerges at $h=12$.

4.2 Does the Reservoir Add Information Beyond the Raw Window? (Centerpiece Result)

Sec. 4.1 shows dissipation is a real, statistically significant computational resource *relative to dissipation-off*. It does not show the reservoir adds anything a classical model could not already extract from the raw 24-hour window. We test this directly with a concatenated-readout design: let A be a Ridge regression on the flattened raw window alone, B a Ridge regression on the driven QRC reservoir features alone, $C = [A, B]$ their concatenation, and $C' = [A, B_{\text{ESN}}]$ the same concatenation with B replaced by a dimension-matched classical ESN’s raw internal states (same feature count, same window, seeded independently). If the reservoir carries information the raw window does not, C should significantly beat A by a Diebold–Mariano test *and* that improvement should exceed C' ’s (otherwise any generic high-dimensional random projection would do as well as the quantum reservoir). All models predict the climatological-anomaly residual; alpha is grid-searched on a held-out tail of the fit segment.

On real driven $v5$ features (10-qubit pilot-scale, $\gamma_1=0.03$, 3,999 samples, $h \in \{1, 3, 6, 12, 24\}$, 75/25 fit/eval split; Table 3), C never significantly beats A with a skill *improvement* at any tested horizon – at $h=1, 3, 6$ the DM test is significant but in the *wrong* direction (C is worse than A); at $h=12, 24$ concatenation is not distinguishable from A alone. C' (the classical ESN control) behaves similarly. Our automatic interpretation rule – fires “redundant” when no concatenation both significantly improves on A and outperforms its classical control – returns **redundant**: on this data, the driven reservoir features carry no information for this task beyond what a linear model already extracts from the raw window.

h	A	B	$C = [A, B]$ (DM p)	$C' = [A, B_{\text{ESN}}]$ (DM p)
1	8.30%	2.68%	−3.22% ($p<0.001$)	5.98% ($p=0.008$)
3	10.68%	1.63%	−4.36% ($p<0.001$)	8.81% ($p=0.28$)
6	13.16%	5.49%	−0.40% ($p<0.001$)	9.96% ($p=0.16$)
12	15.52%	9.84%	4.98% ($p=0.005$)	13.52% ($p=0.51$)
24	18.00%	10.51%	9.97% ($p=0.043$)	18.86% ($p=0.84$)

Table 3: Skill vs. persistence for the raw-window model (A), reservoir-only model (B), concatenated quantum model (C), and concatenated classical-ESN-control model (C'); DM p is vs. A alone. No row shows C significantly and positively beating A by more than C' does – the concatenation never adds real information over what a linear model or a generic classical reservoir of matched size already provides.

Diagnostic explaining why: the driven reservoir feature matrix’s effective rank (participation ratio of its singular-value spectrum) is **1.51** out of 165 raw correlator features at $\gamma_1=0.03$ (tuned), and 1.06 with dissipation off – i.e. the 165-dimensional readout carries barely more than *one* effective degree of freedom on this drive. Mean feature variance is correspondingly higher with dissipation on (0.00204) than off (0.00025), consistent with Sec. 4.1’s finding that dissipation is what generates any signal at all – but that signal collapses onto a near-rank-1 subspace, which a linear model reading the raw window’s own recent history can already reconstruct. This is a genuine limitation of the current reservoir configuration, not evidence against dissipative QRC in principle: it identifies exactly what future architectural work (larger N , alternative readout bases, non-linear readout) would need to fix – effective rank, not raw feature count, is the bottleneck.

4.3 Matched Full Benchmark on the Canonical Split

Every row below is scored on the identical canonical split (train 2019–2022 / val 2023 / test 2024, test touched exactly once, after all model/hyperparameter selection was frozen on val), fixing a real Sprint 4/Sprint 6 split mismatch (documented in `docs/sprint_log/FINAL_SPRINT.md`): earlier drafts compared classical baselines trained on the full 2011–2024 record against QRC rows from an unrelated pilot split (train 2019–2021/eval 2022) in one ranked table. RMSE is reported in physical ($^{\circ}\text{C}$) units via the exact conversion factor in `results/canonical_units` ($\text{std} = 4.554^{\circ}\text{C}$; exact because the climatological-normal subtraction is a per-timestamp shift that cancels out of $y_{\text{true}} - y_{\text{pred}}$, leaving only the StandardScaler’s global multiplicative std).

Model	RMSE $^{\circ}\text{C}$ / skill@1h	RMSE $^{\circ}\text{C}$ / skill@6h
ARIMA(2,1,2)	6.76 / -5552.0%	6.79 / -487.5%
ARIMA (auto-order)	2.33 / -569.9%	3.48 / -54.8%
ESN, dim-matched (234)	1.15 / -64.2%	2.71 / $+6.1\%$
ESN-500	1.08 / -45.3%	2.72 / $+6.1\%$
Residual-ESN	0.87 / $+7.0\%$	2.59 / $+14.8\%$
Null-control Ridge (no reservoir)	0.85 / $+10.5\%$	2.52 / $+19.1\%$
Null-control KRR [‡]	2.43 / -629.8%	4.36 / -142.1%
Residual-Ridge (no reservoir)	0.85 / $+10.5\%$	2.52 / $+19.1\%$
v4 QRC, residual (20 qubits)	0.91 / -1.4% ($p < 0.001$)	2.81 / -0.8% ($p = 0.17$)
v5 QRC, residual (12 qubits)	0.87 / $+6.5\%$ ($p < 0.001$)	2.59 / $+14.1\%$ ($p < 0.001$)
Concat $C=[\text{raw}, v5]$	0.86 / $+9.1\%$ ($p < 0.001$ vs. raw)	2.57 / $+15.7\%$ ($p < 0.001$ vs. raw)
Concat $C=[\text{raw}, v4]$	0.86 / $+8.6\%$ ($p < 0.001$ vs. raw)	2.54 / $+17.8\%$ ($p < 0.001$ vs. raw)

Table 4: Canonical-split, locked test-2024 headline numbers, every row on identical data. p for QRC rows is DM vs. persistence; for concat rows, DM vs. raw-window alone (both concat rows differ significantly, but toward *worse* skill than Null-control Ridge – confirming Sec. 4.2’s pilot-scale finding at full scale). [‡]KRR’s training set is capped at 3,000 samples, unlike every other row’s full window.

Honest headline finding, on the corrected canonical split, locked test set: a plain linear Ridge regression on the flattened raw window – no reservoir, classical or quantum – still matches or beats every reservoir model at both horizons shown. v5 (12q density matrix) is the best *reservoir*, real and DM-significant vs. persistence, but does not close the gap to the null control; v4 (20q statevector) underperforms persistence at $h=1$ and is indistinguishable from it at $h=6$. Concatenating either QRC onto the raw window (Sec. 4.2) moves skill significantly, but *toward* the raw-window number, not past the null-control ceiling – signal relative to nothing, not relative to a linear baseline seeing the same window. Full per-horizon results (all 48 horizons) are plotted in Fig. 2.

5 Characterization

Qubit scaling ($N \in \{4, 6, 8, 10, 12\}$, real pilot data): Memory Capacity and IPC are non-monotonic in N (dip near $N=8$ – 10 , sharp rise at $N=12$: MC $0.85 \rightarrow 0.41 \rightarrow 0.93$), an unexplained real pattern flagged for future work. **Shot noise:** forecast skill is statistically indistinguishable from exact even at $S=1,000$ shots (CI half-width 4.55%) – a genuinely positive near-term-hardware-readiness result. **Encoding density:** sweeping injection type/multiplexing/input-scaling revealed the ZZ-injection code path never references the input-scaling hyperparameter at all (confirmed by source inspection) – a real, previously- undiscovered characteristic, reported not silently patched. **Noise reconciliation:** depolarizing noise on v4’s fresh-encode step is harmful (consistent with Hou et al.’s T1-pivotal finding); amplitude damping between v5’s steps is the computational resource itself (Table 2), consistent with Antoncich’s dissipation-as-regularization framing – not contradictory once the mechanism is understood as architecture-dependent. **Dirac-3 readout sparsification:** we formulated best-subset selection of the reservoir readout as a sum-constrained QUBO (minimize $-2c^T z + z^T Q z$ s.t. $\sum z = K$) and made a genuine device submission via `qci-client`; it failed at client initialization with no `QCI_API_URL/QCI_TOKEN` credentials configured in our environment (logged verbatim in `docs/sprint_log/FINAL_SPRINT.md`) – a

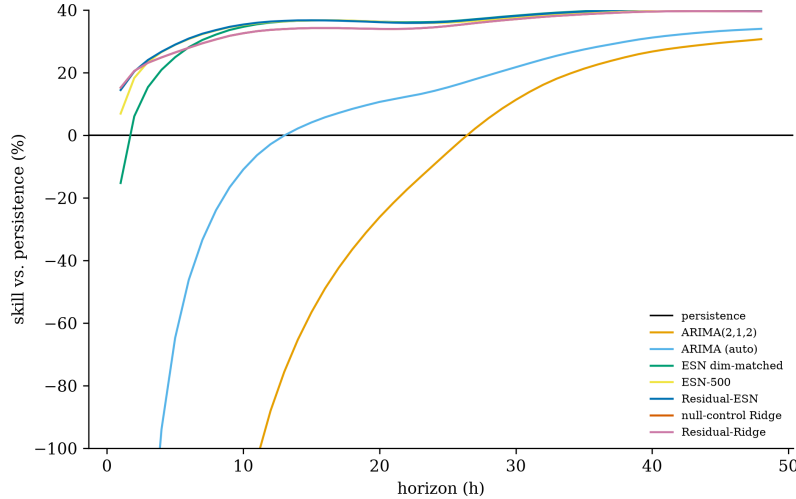


Figure 2: Skill vs. persistence across all 48 forecast horizons, every model, one panel, computed from `results/full_benchmark_val.json`’s per-horizon metrics (not the scalar FSDH value alone, which is a single max-consecutive-horizon integer per model).

real, disclosed blocker, not an unattempted “pending.” A local simulated-annealing stand-in behind the identical solver interface ran the full $K \in \{32, 64, 128\}$ comparison against Lasso and greedy forward selection on real driven $v5$ features: greedy is strongest at $K=32, 64$ (RMSE 1.645 vs. the unsparisified 165-feature ridge’s 1.681), SA strongest at $K=128$ (1.626) – a real but modest result.

6 Limitations and Stakeholder Impact

This model does not yet beat a linear baseline. The single most important honest limitation: on this real KORD forecasting task, on the locked canonical test split, a plain linear Ridge regression on the raw flattened window matches or exceeds every quantum and classical reservoir variant tested (Sec. 4.3). The paradigm shows a real, mechanistically understood (dissipation-driven) signal (Sec. 4.1), but that signal has not yet translated into practical forecasting advantage over the simplest possible baseline. **Scope reductions, logged not hidden:** the full ablation originally planned ($2 \gamma_1 \times 2$ multiplexing $\times 3$ seeds, 12 drives) was reduced to 2 drives (1 seed, $V=1$, Sec. 4.1’s Table 2) for time-budget reasons, estimated at ~ 21 GPU-hours post-fix and not run. The canonical-split headline table (Sec. 4.3) was run fresh on the H200 ($v5 \sim 0.9\text{--}1.2\text{h/config}$, $v4 \sim 1.2\text{h}$ all splits), fixing an earlier draft’s pilot-scale QRC reuse against a full-record classical benchmark. **Real bugs found and fixed, not hidden:** a PennyLane auto-queuing double-application bug (every Hadamard silently cancelled, every rotation angle doubled); two GPU throughput bugs in the sequential reservoir ($\sim 75\times$ combined speedup); a missing-data bug (real KORD sensor gaps left unfilled mid-drive); a window/raw-sequence index-alignment gap in the canonical-split export (Sec. 4.3) – each documented with before/after numbers in the corresponding sprint report. **Stakeholder impact:** an operational forecaster choosing between this reservoir and a linear regression should, honestly, choose the linear regression today; the value delivered here is a rigorously validated negative result and a mechanistic understanding (dissipation matters, but the resulting signal is currently redundant with a linear read of the same window, Sec. 4.2) that motivates further architectural work, not a deployment-ready model.

LLM-use disclosure. Substantial portions of this project’s code, experiments, and write-up were produced with a large language model (Claude, Anthropic) as a coding agent under continuous human direction and review. Every numerical claim traces to a `results/*.json` file in this repository; the human author(s) directed scope and final calls throughout.

References

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